Short Workshop: Introduction to Meta-Analysis and Systematic Reviews
June 25 – June 26, 2015

Instructors:
Jessica Gurevitch (Stony Brook University, State University of New York)
Marc Lajeunesse (University of South Florida)

A 1.5 day workshop will be held before the Evolution meeting in Guarujá, Brazil, starting on the afternoon of June 25. The cost is $60 U.S. per participant. Some scholarships may be available. Applications and registration details are available at:

http://lajeunesse.myweb.usf.edu/ASN_workshop_2015.html

Participation is limited to 24 people. Sponsored by the *American Society of Naturalists*.

Introduction/Objectives
Synopsis: Participants will become familiar with what meta-analysis is, where it comes from, some examples of how it has been used in ecology and evolution, and its major strengths, weaknesses, advantages and limitations as a tool for the quantitative summary of research results. They will be introduced to the formal methods for carrying out a systematic review, which may (or may not) be combined with a meta-analysis, and to the problems and issues involved in carrying out systematic reviews. They will become familiar with carrying out the calculations involved in typical ecological meta-analyses, and learn how to interpret the results. We will look very briefly at some of the areas of current development in meta-analysis in ecology, evolution, and conservation, including integrating phylogenetic information in analyses and imputation methods. We also introduce criteria for evaluating and critiquing the results of meta-analyses in the literature.

Workshop participants should have a general background in statistics (anova, regression, etc.) and graduate-level understanding of experimental design, ecology, evolution and/or conservation research. No real familiarity with meta-analysis is necessary.

Format of the course:
Lectures, demonstrations of methods, hands-on computational experience with ample opportunities for questions and feedback, group discussions and presentations, and peer critiques.

Prerequisites: An introductory statistics course covering up to regression and ANOVA. Background in ecology and evolutionary biology. Familiarity with experimental design and more advanced statistics and probability desirable but not necessary.

Software:
OpenMEE (open meta-analysis software for ecology and evolution). Freely available at: http://www.cebm.brown.edu/open_mee. Familiarity and experience with *R* is not necessary but will prove useful.

Individual Assignment: *Prior to the course, read:*

SELECTED CHAPTERS FROM Koricheva, Gurevitch and Mengersen, 2013, Handbook of Meta-analysis in Ecology and Evolution (to be added)

Participation is limited to 24 people.

About the Instructors:
Gurevitch and Lajeunesse have taught meta-analysis short courses and workshops in many different venues worldwide, both alone and together.

Jessica Gurevitch is a Professor in the Department of Ecology and Evolution at Stony Brook University, New York. Her research interests are in plant ecology and statistical applications in ecology. Major current interests are plant demography in disturbed environments, the ecology of biological invasions, and systematic reviews and meta-analysis in ecological research. She introduced contemporary quantitative research synthesis and meta-analysis to the fields of ecology and evolution, changing the way scientists in these fields conceptualize and review scientific data. She received a B.S. in Biological Sciences with a concentration in Ecology, Evolution & Systematics from Cornell University and a Ph.D. in Ecology and Evolutionary Biology from the University of Arizona. In 2011 she was elected a Fellow of the American Association for the Advancement of Science (AAAS) in the Biological Sciences, a Fellow of the Ecological Society of America, and is an elected member of of the Society for Research Synthesis Methodology.

Marc Lajeunesse is an Assistant Professor at the University of South Florida, Tampa. His research interests include developing metrics and statistics for ecological meta-analysis, addressing ecological and evolutionary questions about parasites and herbivores using these tools, and exploring the evolutionary and chemical basis of host specificity using laboratory experiments with herbivorous mites and their diverse host plants. He has developed and introduced methods for phylogenetic meta-analysis and for imputation for missing data in ecological meta-analysis. He received a B.Sc. and M.Sc in Biology from Carleton University (Ottawa, Canada), a Ph.D. in Ecology from Cornell University, and was a postdoctoral fellow at the National Evolutionary Synthesis Center (NESCent) at Duke University.